Efficient Ensemble Model for Facial Expression Recognition

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ABSTRACT

As the trend in deep learning currently revolves around scaling models for better performance on different tasks, such as facial expression recognition (FER), exploring alternative methods for achieving great performance while using low computation is of high interest. Hence, this study focuses on investigating the performance in terms of accuracy and efficiency of FER when combining small models utilizing ensemble methods with those of a large model. Our results show that ensembling three smaller models with simple averaging was just 1.61% short of the larger model in terms of accuracy, while the total training time of the smaller models combined was less than half of that required for the larger model. These findings highlight the potential of using multi-class ensembles to enhance performance in facial expression recognition tasks while maintaining low computational costs. Notebooks and datasets are all available in this GitHub repository.

Keywords: Facial expression recognition, FER-2013, Ensemble deep learning, Compact models ensembling, Efficient architectures, one-vs-rest binary classifiers

1 Introduction

In recent years, facial expression recognition (FER) has gained lots of attention. This has primarily to do with its potential applications in various fields, such as evaluating customer service satisfaction, human-computer interaction systems, and the development of accessibility features for disability-inclusive systems [6]. Furthermore, the field has experienced an upswing in terms of model performance, especially in accuracy. Primarily, this is due to the models becoming more complex as the number of parameters increases, which is commonly known as scaling. Consequently, this creates a need for more computational power as the models now have more parameters to update during training, which in turn leads to a current tradeoff between accuracy and efficiency [1], [7], [8], and [10]. Hence, it is well motivated to seek alternative methods of scaling to increase model performance on FER.

As computational power is neither easy nor affordable to lay hands on, we aim to compare a large model's accuracy and efficiency to those of smaller models that utilize ensembling techniques. Hence, our vision is to investigate the potential for achieving comparably great accuracy to that of a larger model while using significantly less computation time by utilizing and optimizing ensembling techniques with smaller models.

2 Related Works

Goodfellow et al. (2013) made Facial Expression Recognition (FER) a Kaggle competition to persuade analysts to form more advanced FER frameworks. Computer vision investigations on FER have progressed, with profound learning models creating state-of-the-art results. The exactness and effectiveness of these expression recognition frameworks have been revolutionized by using convolutional neural networks (CNNs). This area digs into the assorted deep-learning approaches recommended within the writing, emphasizing their particular strategies, focal points, and restrictions.

Standard CNN architectures like AlexNet and VGG-Face were originally developed for object recognition but have proven highly effective for tasks such as facial expression recognition (FER). These models showcase the superiority of deep-learned features over traditional handcrafted features, which has encouraged the development of more specialized and sophisticated architectures. Among these, notable advancements in performance have been reported: Tang (2013) achieved a significant 71.2% accuracy using an SVM's primal objective to enhance the model's capability to distinguish subtle facial expressions [14]. Similarly, Khaireddin et al. (2021) reached a state-of-the-art accuracy of 73.28% by meticulously fine-tuning the VGGNet architecture and its hyperparameters, demonstrating the potential of deep learning models when optimized for specific tasks like FER [6].

Recent research has focused on tailoring common models to capture the subtle nuances of human emotions more accurately. Among these adaptations are depthwise separable convolutions, inception modules, and attention mechanisms, which collectively improve the computational efficiency and learning capacity of FER frameworks. Notable implementations include Liu et al.'s (2016) work, where individual CNNs were optimized and then amalgamated, achieving a best single-network accuracy of 62.44% [12]. Similarly, Minae et al. (2019) utilized an attentional convolutional setup in an end-to-end deep learning system to reach a 70.02% accuracy rate [15]. Tang et al. (2013) furthered this by replacing the softmax layer with a support vector machine in their neural configuration, resulting in a classification accuracy of 71.2% [16]. A comparative analysis by Khaireddin et al. (2021) evaluated VGG, Initiation, and ResNet architectures, finding that VGG led with the highest precision at 72.7%, closely followed by ResNet at 72.4% and Initiation at 71.6%, showcasing the efficacy of these evolved architectures in understanding and interpreting human emotions [6].

Deeper CNNs require greater computational resources and are more difficult to train due to their complex architecture and deeper layers. Several studies have suggested employing an ensemble technique that focuses on developing a compact, efficient network architecture as a solution to this problem. This technique involves assembling subnet models from popular CNN architectures such as VGGNet, ResNet, and AlexNet with no more than 10 layers [12], [13]. The method is to train each of these subnets separately on the training set first, and then combine them by removing their output layers. The accuracy is then increased even higher by using a Support Vector Machine (SVM) classifier. The accuracy of the ensemble is 65.03% without the SVM and increases to about 70% with the SVM.

While these prior studies employ similar techniques to our method, they still use CNN architectures that are considered large. Our method, on the other hand, attempts ensembling with compact models [4], [5], [11], and introduces a novel approach to the ensembling technique by utilizing methods such as binary one-vs-rest classification.

3 Ensemble Deep Learning for Facial Expression Recognition

We introduce an ensemble model (Ensemble A, Figure 1) for facial expression recognition, aiming to achieve high accuracy by combining smaller models compared to EfficientNet B3, a large-scale model, and human accuracy levels. To select the most efficient models, we use Grad-CAM to visualize the focus areas of networks from various unique, compact architecture families. This allowed us to assess how each model extracted features. Ultimately, we narrowed down our selection to three models, ShuffleNet[11], MobileNet[4], and SqueezeNet[5], which demonstrated complementary focus areas (Figure 4). In the training phase, different bootstrap samples were used to ensure that each network learned different features from the FER 2013 dataset.

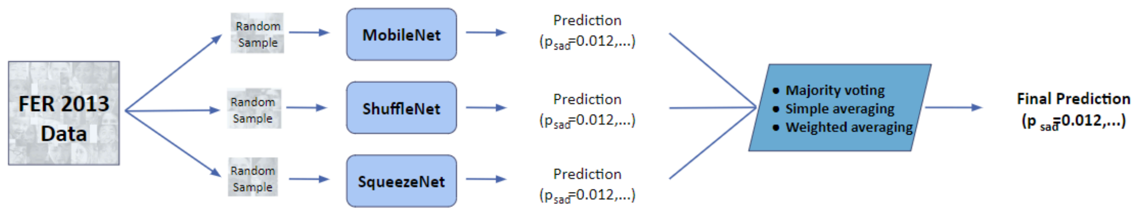

Figure 1: Ensemble A architecture with all multi-class classifiers

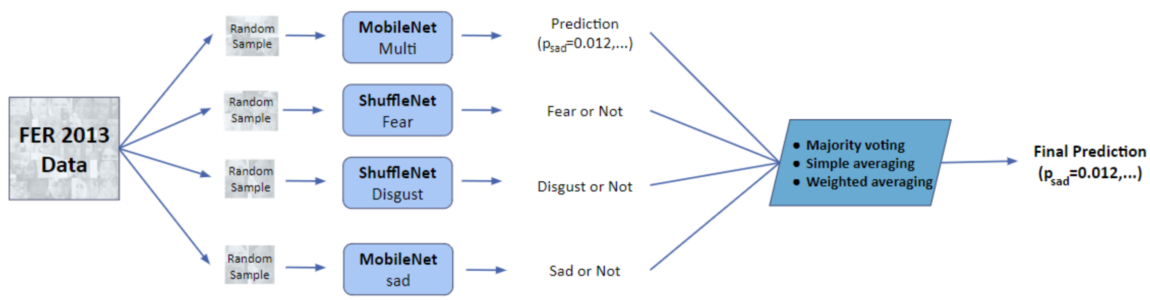

Figure 2: Ensemble B architecture with binary classifiers

Our preliminary results (Figure 3) showed that these models consistently have lower recall on three emotions: fear, sadness, and disgust. So, we propose another ensemble architecture (Ensemble B, Figure 2) that has one multi-classifier network and three binary-classifier networks for detecting these emotions. For each expression, we trained the architecture with the highest recall as a binary classifier to improve the accuracy.

To aggregate the results from each network in both Ensembles A and B, we compare majority voting (Eq. 1), simple averaging (Eq. 2), and weighted averaging (Eq. 3).

$$\hat{y}_{ensemble} = \underset{c}{\operatorname{argmax}} \sum_{b=1}^{B} I(\hat{y}_b = c) \qquad \text{(Eq. 1)}$$

$$\hat{y}_{ensemble} = \underset{c}{\operatorname{argmax}} \left(\frac{1}{B} \sum_{b=1}^{B} \hat{f}_{b,c} \right) \qquad \text{(Eq. 2)}$$

$$\hat{y}_{ensemble} = \underset{c}{\operatorname{argmax}}\left(\sum_{b=1}^{B} w_{b,c}\, \hat{f}_{b,c}\right), \qquad \sum_{b=1}^{B} w_{b,c} = 1 \qquad \text{(Eq. 3)}$$

Where c is the facial expression class index, B is the total number of networks in the ensemble, $I(.)$ is the indicator function, which equals 1 if the condition inside the function is true and 0 otherwise, $\hat{y}_b$ is the predicted label from model b, $\hat{f}_{b,c}$ is the predicted probability for class c from model b, and $w_{b,c}$ is the normalized weight of prediction probabilities for class c from model b. Note that each model will have different weighting factors, and each class in every model can also have different weighting factors to account for each model's strengths and weaknesses.

For the binary classifiers, the prediction probability of the binary problem will be converted into one-hot multi-class probabilities before performing simple averaging and weighted averaging. For the weighted averaging, the corresponding expressions (fear, disgust, and sadness) will have a value according to the binary classifiers' prediction probability, and the remaining expressions will have all zero values. For simple averaging, values of 1 will be used instead of the actual binary classifiers' prediction probability. For example, when the binary classifier for disgust predicts a probability of 0.78, it will be converted into [0, 0.78, 0, …, 0] before performing the weighted averaging or into [0, 1, 0, …, 0] before performing the simple averaging.

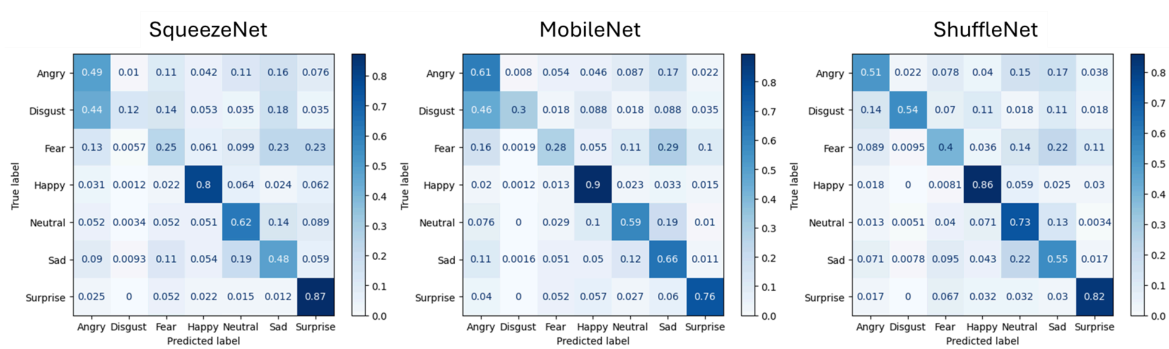

Figure 3: Confusion matrix from the individual models

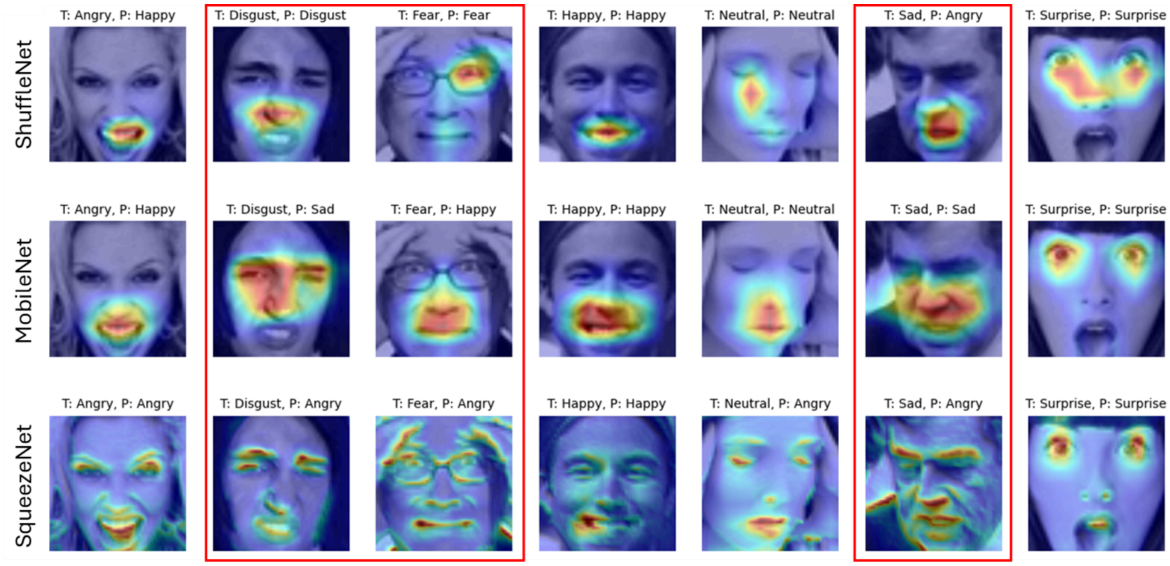

Figure 4: GradCAM visualizations from the individual models

4 Experiment

4.1 Experimental Setup

To evaluate the performance of our proposed method, we use the facial expression recognition FER-2013 dataset as the benchmark [3]. FER-2013 is a dataset of centered grayscale images of faces consisting of 7 classes of basic facial expressions (Angry, Disgust, Fear, Happy, Neutral, Sadness, and Surprise), with around 28.7k training images, 3.6k validation images, and 3.6k test images.

Several transformations were done to the images as described by each architecture's implementation paper [4], [5], [11], to ensure that the input images are in accordance with the pre-trained model's expected input. We also perform common data augmentation techniques such as random horizontal flip, random rotation, and random erasing (cutout) on the training images to avoid overfitting [9]. These training images are then bootstrapped and randomly sampled with the same size as the total training subset but with replacements for each of the networks in the ensemble.

Weighting factors for each class in each model's prediction were set as follows; For each class, we assign a higher weight (0.4) to the model with the highest recall on that corresponding class during the individual preliminary experiment and divide equally among the remaining models until it all adds up to 1. However, for ensemble B, which has binary classifier members, weights are only assigned to the fear, disgust, and sadness classes.

For training, we use a uniform batch size of 48 and train the model on an NVIDIA L4 GPU device for a maximum of 15 epochs. We use each architecture's pre-trained weight as an initialization point and then retrain all of the layers with our training images. We will compare the performance of our proposed ensemble of smaller models to a relatively large model, EfficientNet B3. We reimplement all of the models and train them on the same hardware to calculate efficiency in terms of time spent training accurately under the same computational resource constraints.

4.2 Experimental Results

Table 1: Performance evaluations on FER-2013 test set

Model	Aggregation	Accuracy (%)	Training time	Params (M)	MACs (G)
Ensemble A	Majority vote	66.56	**35m 28s**	**10.30**	**1.08**
	Simple average	**67.01**			
	Weighted average	66.95			
Ensemble B	Majority vote	60.99	72m 12s	19.14	1.64
	Simple average	60.30			
	Weighted average	56.03			
EfficientNet B3		**68.62**	85m 48s	10.71	1.93
Human accuracy		65.00	-	-	-

We compared the performance of two ensemble models, Ensemble A (Figure 1) and Ensemble B (Figure 2), and EfficientNet B3 as a benchmark large model. We compared each model's accuracy, training time, number of parameters, and multiply-accumulate operations (MACs). Additionally, we compared the ensemble models' performance to human-level accuracy.

Ensemble A had the overall highest performance across different aggregation methods. Ensemble A reaches the highest accuracy of 67.01% with the simple averaging method. The majority voting and weighted averaging methods, resulted in slightly lower accuracy of 66.56% and 66.95%, respectively. Ensemble B, however, showed a lower performance compared to Ensemble A. The majority vote method achieved an accuracy of 60.99%, while the simple average method recorded 60.30%. The weighted average method resulted in a significantly lower accuracy of 56.03%. Our large model, EfficientNet B3, had the highest accuracy of 68.62% among all.

However, the training time for Ensemble A was only 35 minutes and 28 seconds, nearly twice as fast as Ensemble B (72 minutes) and the larger model, EfficientNet B3 (85 minutes). Considering its high accuracy of 67.01%, Ensemble A proves to be a strong competitor to the larger EfficientNet B3 model. Moreover, both Ensemble A and EfficientNet B3 outperform the human accuracy of 65% on the FER-2013 dataset.

Since the accuracy for certain emotions (Fear, Disgust, and Sadness) was significantly lower, we proposed the idea of using binary classification and expected to reach a higher accuracy. However, as shown in Table 1, the accuracy decreased. This may happen for two possible reasons. Firstly, Ensemble B becomes overconfident in detecting these emotions due to its binary classification approach, which means it is less generalized and fails to detect all emotions correctly. Secondly, Ensemble B has higher complexity, with 19.14 million parameters and 1.64 giga MACs, compared to Ensemble A's 10.30 million parameters and 1.08 giga MACs. This can cause overfitting, where the model performs well on training data but poorly on unseen test data.

5 Conclusion

This research compares the effectiveness and efficiency of two types of ensembles, Ensemble A with all multi-class classifiers and Ensemble B utilizing a one-vs-rest binary classifier for FER, against EfficientNet B3, a large model.

From our findings, we observed that Ensemble A, which used three multi-class emotion detection models, attained similar accuracy to our larger model, EfficientNet B3, using significantly fewer computational resources and only half the training time. These findings suggest that it is possible to develop a high-performance face recognition system with lower computational budgets by employing ensemble techniques instead of creating larger, more complex networks.

Furthermore, we discovered that the addition of the binary classifier in the proposed system did not increase its performance but led to decreased accuracy. This degradation in precision could be attributed to over-training as well as the inclusion of more parameters and multiply-accumulate operations (MACs) compared to Ensemble A. Nevertheless, further investigations may focus on tuning the number of binary classifiers and multi-predictors for better accuracies and training times, while other future works should look into more aggregation methods and determine optimum weights for weighted averaging

REFERENCES

[1] Dosovitskiy, Alexey, et al. "An image is worth 16x16 words: Transformers for image recognition at scale." arXiv:2010.11929, 2020.

[2] Ganaie, Mudasir A., et al. "Ensemble deep learning: A review." Engineering Applications of Artificial Intelligence 115 (2022): 105151.

[3] Goodfellow, Ian J., et al. "Challenges in representation learning: A report on three machine learning contests." In ICONIP. Proceedings, Part III 20. Springer berlin heidelberg, 2013.

[4] Howard, Andrew G., et al. "Mobilenets: Efficient convolutional neural networks for mobile vision applications." arXiv:1704.04861, 2017.

[5] Iandola, Forrest N., et al. "SqueezeNet: AlexNet-level accuracy with 50x fewer parameters and< 0.5 MB model size." arXiv:1602.07360, 2016.

[6] Khaireddin, Yousif, and Zhuofa Chen. "Facial emotion recognition: State of the art performance on FER2013." arXiv:2105.03588, 2021.

[7] Liu, Ze, et al. "Swin transformer: Hierarchical vision transformer using shifted windows." Proceedings of the IEEE/CVF ICCV. 2021.

[8] Pham, Luan, The Huynh Vu, and Tuan Anh Tran. "Facial expression recognition using residual masking network." In ICPR. IEEE, 2021.

[9] Shorten, Connor, and Taghi M. Khoshgoftaar. "A survey on image data augmentation for deep learning." Journal of big data 6.1 (2019): 1-48.

[10] Tan, Mingxing, and Quoc Le. "Efficientnet: Rethinking model scaling for convolutional neural networks." In ICML. PMLR, 2019.

[11]Zhang, Xiangyu, et al. "Shufflenet: An extremely efficient convolutional neural network for mobile devices." Proceedings of the IEEE CVPR. arxiv:1707.01083, 2018.

[12] K. Liu, M. Zhang and Z. Pan, "Facial Expression Recognition with CNN Ensemble," 2016 International Conference on Cyberworlds (CW), Chongqing, China, 2016, pp. 163-166, doi: 10.1109/CW.2016.34.

[13] C. Jia, C. L. Li and Z. Ying, "Facial expression recognition based on the ensemble learning of CNNs," 2020 IEEE International Conference on Signal Processing, Communications and Computing (ICSPCC), Macau, China, 2020, pp. 1-5, doi: 10.1109/ICSPCC50002.2020.9259543.

[14] Y. Tang, “Deep Learning using Support Vector Machines,” in International Conference on Machine Learning (ICML) Workshops, 2013.

[15] S. Minae and A. Abdolrashidi, “Deep-emotion: facial expression recognition using attentional convolutional network,” arXiv. 2019, doi: 10.3390/s21093046.

[16] Y. Tang, “Deep learning using linear support vector machines,” arXiv, vol. 1306.0239, 2013.

A APPENDICES

A.1 Experiments with Individual Models

Table A1: Classification Metrics of EfficientNet on FER-2013 Dataset

Emotion	Precision	Recall	F1-Score	Frequency
Angry	0.61	0.56	0.58	497
Disgust	0.58	0.72	0.64	57
Fear	0.61	0.46	0.53	526
Happy	0.81	0.92	0.86	869
Neutral	0.6	0.7	0.64	593
Sad	0.59	0.53	0.56	645
Surprise	0.78	0.78	0.78	402
Total/Average	0.67	0.68	0.67	3589

Table A2: Classification Metrics of ShuffleNet on FER-2013 Dataset

Emotion	Precision	Recall	F1-Score	Frequency
Angry	0.66	0.51	0.57	497
Disgust	0.56	0.54	0.55	57
Fear	0.56	0.40	0.47	526
Happy	0.85	0.86	0.86	869
Neutral	0.55	0.73	0.63	593
Sad	0.53	0.55	0.54	645
Surprise	0.74	0.82	0.78	402
Total/Average	0.66	0.66	0.65	3589

Table A3: Classification Metrics of MobileNet on FER-2013 Dataset

Emotion	Precision	Recall	F1-Score	Frequency
Angry	0.54	0.61	0.58	497
Disgust	0.71	0.30	0.42	57
Fear	0.58	0.28	0.38	526
Happy	0.82	0.90	0.85	869
Neutral	0.63	0.59	0.61	593
Sad	0.51	0.66	0.57	645
Surprise	0.77	0.76	0.76	402
Total/Average	0.65	0.65	0.64	3589

Table A4: Classification Metrics of SqueezeNet on FER-2013 Dataset

Emotion	Precision	Recall	F1-Score	Frequency
Angry	0.53	0.49	0.51	497
Disgust	0.29	0.12	0.17	57
Fear	0.38	0.25	0.30	526
Happy	0.84	0.80	0.82	869
Neutral	0.55	0.62	0.58	593
Sad	0.49	0.48	0.49	645
Surprise	0.53	0.87	0.66	402
Total/Average	0.58	0.58	0.57	3589

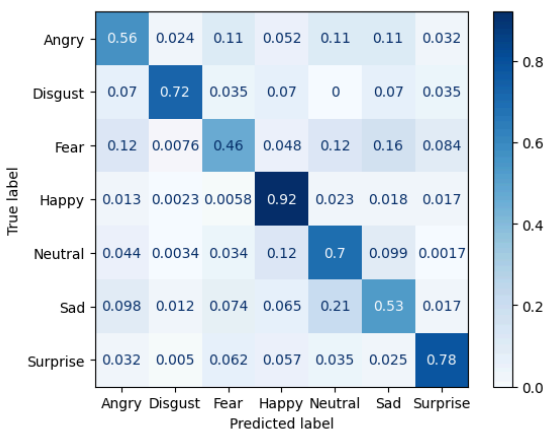

Figure A1: Confusion Matrix of EfficientNet on FER-2013 Dataset

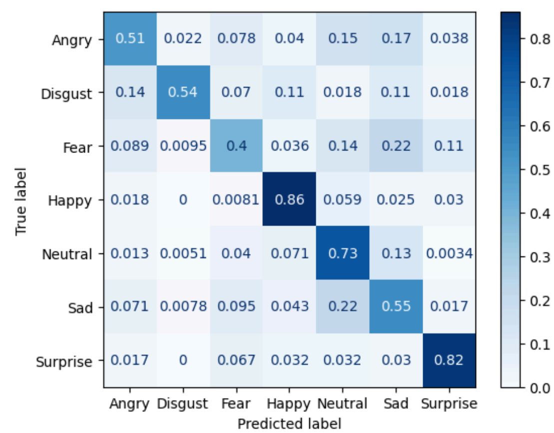

Figure A2: Confusion Matrix of ShuffleNet on FER-2013 Dataset

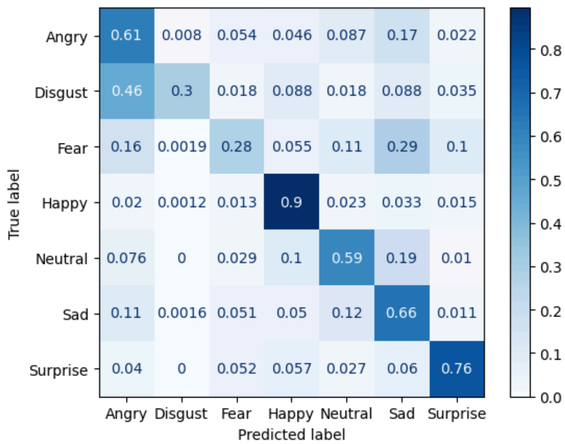

Figure A3: Confusion Matrix of MobileNet on FER-2013 Dataset

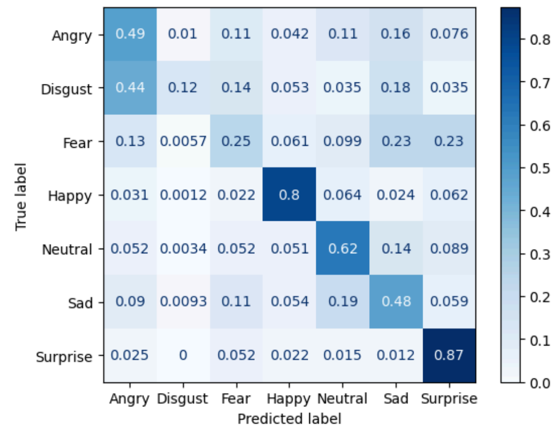

Figure A4: Confusion Matrix of SqueezeNet on FER-2013 Dataset